

Homework 3

STAT 300 (Fall 2024)

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Clearly label your answers to each question and each sub-question. Your answers MUST be uploaded to Canvas as a <HWx_Yourfirstname.nb.html> file by the deadline.

(1). The *RailsTrails.csv* data include information on a random sample of 104 homes that were sold in 2007 in the city of Northampton, MA. The goal is to see if the home's proximity to a biking trail would relate to the selling price. For this you will be using variables *Adj2007* (given in 1000's of dollars) which is the sold price in 2007 adjusted for inflation and *Distance* which is the distance in miles to the nearest entry point of a trail.

- (a). Create an appropriate graphs to see the relationship between the sold price and the distance to a trail.
- (b). Comment on your observations.
- (c). Use R to obtain the least squares regression model.
- (d). Write down the least squares regression model.
- (e). Interpret the slope and the intercept in the context of this problem.
- (f). What is the regression standard error?
- (g). Check the validity of the regression assumptions by making appropriate graphs and commenting on each of the assumptions. When you comment on the validity of the assumptions, you have to clearly state why or why not a certain assumption is violated.

(2). For this problem you will be again using the *RailsTrails.csv* data. Now you are going to see if price (*Adj2007*) is associated with the size of the house (*SquareFeet*-given in 1000's).

- (a). Create an appropriate graphs to see the relationship between the sold price and the distance to a trail.
- (b). Comment on your observations. Does this align with what you would have expected to be?
- (c). Use R to obtain the least squares regression model.
- (d). Write down the least squares regression model.
- (e). Interpret the slope in the context of the problem.
- (f). Check the validity of the regression assumptions by making appropriate graphs and commenting on each of the assumptions. When you comment on the validity of the assumptions, you have to clearly state why or why not a certain assumption is violated.

(3). A capacitor was charged with a 9-volt battery and then a voltmeter recorded the the voltage as the capacitor was discharged. Measurements were taken every 0.02 second. The data is in *Volt.csv* file.

- (a). Make a scatterplot with voltage on the y-axis and time on the x-axis. Comment on what your observations.
- (b). Regardless, fit a linear model and obtain the fitted regression model.
- (c). Write the equation of the least squares regression model.

- (d). Create a plot of standardized residuals vs. fitted values plot. What does the plot say about fitting a linear model to predict voltage from time?
- (e). Transform voltage with log transformation and plot log(voltage) vs. time. Comment on what you see.
- (f). Regress log(voltage) on time and write the equation.
- (g). Now make a plot of standardized residuals vs. fitted values plot for the model you obtain in part (f). Comment on your observations.
- (h). Using the fitted model to the transformed data, predict the voltage when the elapsed time is 0.6 sec.
- (i). How close is your prediction to the observed response?

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#To find the observed response when time =0.6, you may adapt a code like below.
#The object name I have given for the data frame is volt
Row0.6<-which(volt$Time==0.6) #This should give you the entry number in which time is equal to 0.6
volt$Voltage[Row0.6]
```

- (4) . Many different interest groups such as the lumber industry, ecologists, and foresters benefit from being able to predict the volume of a tree just by knowing its diameter. One classic data set (*shortleaf.txt*) reported by C. Bruce and F. X. Schumacher in 1935 concerned the diameter (in inches) and volume (in cubic feet) of 70 shortleaf pines.

A researcher is interested in learning about the relationship between the diameter and volume of shortleaf pines.

- (a). Identify the response variable and explanatory variable for the problem
- (b). Draw a scatter plot to show how volume of a tree and its diameter are associated. Comment on your observations.
- (c). Fit a regression line for the problem, write down the estimated equation (define any terms you might have used), and overlay the estimated line on the scatter plot.
- (d). Obtain the diagnostics for the fitted model in part (c). Clearly state your observations. Provide all the outputs you used.
- (e). Identify (i) the point with highest residual (studentized residual), (ii) the point with highest leverage, and (iii) the point with highest Cook's distance. Suppose a friend of the researcher suggested that there is an influential point in the data, and should be investigated. Do you agree with this comment? Explain your reasoning.

```
#identifying the point with highest residual you can use a code line similar to below.
#shortleafData is the name of my data object
#I have previously saved the studentized residuals into an object named stud.residuals
#abs() function gives the absolute value of argument inside the function and
#max() function gives the maximum of the argument passed inside the function
#== is a logical function that checks left-hand side is equal to right-hand side
shortleafData[abs(stud.residuals)==max(abs(stud.residuals)),]
```